

project-one

Clean, consistent scoping documentation for the **storage capacity management & deviation detection** project: an AI / agentic (LangGraph) system that analyzes filesystem and storage configurations across a heterogeneous multi-OS fleet (AIX, Linux, all distributions, Windows Server), at least 500 servers, to serve five objectives: **01** deviation detection vs standards, **02** optimization (useless / obsolete / misplaced), **03** reliability & full-filesystem prevention, **04** operational efficiency & forecasting, **05** risk reduction & compliance.

This folder holds the project's scoping documentation: a single canonical data model, one consistent terminology, English throughout.

Canonical data model (in one line)

Each collection run produces **one envelope file** (`{ snapshot_id, generated_at, host_count, hosts: [ ... ] }`) where every `hosts[]` element is one server's document in a **single, OS-agnostic schema** (`filesystems[] + storage_topology + paging[]`), all sizes in **exact bytes**, non-applicable fields set to explicit `null`. Run identifiers live on the envelope; each host carries only its `scan_timestamp`.

Documents (`docs/`)

Read in this order:

1. [Project Overview](#): purpose, scope, the five objectives, approach, process, deliverables, key locked decisions, glossary. **Start here.**
2. [Business Goals](#): B-01...B-13 with areas, value, mapped objectives, and KPIs.
3. [Requirements](#): functional (FR-01...FR-20) and non-functional (NFR-01...NFR-17).
4. [Risk Register](#): RSK-01...RSK-23 with mitigations, plus the top-risk responses.
5. [Data Collection Specification](#): the authoritative data contract: the OS-agnostic schema, per-level field tables, objective coverage, excluded derived fields, and worked JSON examples.
6. [Data Collection Request](#): the ready-to-send message to the data / collection team (cover note + Appendix A schema + Appendix B field rationale).
7. [LangGraph graph](#): how the LLM layer is structured as a LangGraph graph: state, nodes, routing, and the wiring. Diagram (PNG + SVG + Mermaid source) in [docs/diagrams/](#).

Next steps (out of scope for this set)

Machine-validatable JSON Schema file, per-OS reference standards / golden images, OS-by-OS field-applicability matrix, field-by-field rationale + Ansible collection roles.

Status: **scoping / pre-implementation** (no application code yet), 2026-06-15.